Native Optimal-Completion Fibers, Real-Time Pushdown Recognition, and Protected Paired-Session Comparisons

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Abstract

For each finite simplicial complex with a prescribed facet presentation, we give an explicit completion-total integer counting source whose true shortest unit-cost completion fiber has one point per facet and whose selected action-support complex is exactly the prescribed complex. Its active integral difference lattice is primitive, so the periodic optimal-fiber quotient has a free comparison group with its natural displacement marking. A bounded-length radix encoding replaces the entire vector source by scalar integer action weights without changing a single shortest Parikh completion. The resulting infinite zero-return language is accepted by a deterministic, real-time, one-stack automaton. Under a separately specified paired-rooted-session interface, another real-time one-stack service reads genuine source words and maintains the freely reduced marked comparison word without adding input letters. We prove exact impossibility results for a single uninterrupted rooted shortest trace and for irreversible prefix-exclusive service selection, keeping them separate from unrestricted single-start dual-output realization. The construction and its limits are given explicitly on a seven-letter source.

Keywords. shortest completion; unit-cost counting source; optimal Parikh fiber; deterministic pushdown automaton; periodic nerve; protected session comparison.

1 Main contributions and mathematical scope

The article establishes three linked assertions. First, an explicit positive-kernel, unit-cost source realizes a prescribed simplicial support complex *and* a primitive optimal completion fiber at the same target; the associated marked quotient is calculated using the earlier fiber theory. Second, an integral radix map preserves that entire optimum literally while producing a scalar source with a global real-time pushdown zero-return recognizer. Third, a separately selected, clocked interface compares pairs of genuine rooted shortest source sessions on one stack and computes freely reduced comparison words without extending the source alphabet. The article also proves two precise restrictions on interpreting multiple rooted sessions as one directed computation or routing between services from a dedicated finite input prefix.

These are explicit *interface-dependent* assertions. The source is changed by the radix map, only a stated shortest target fiber is preserved, and the two automata services have different entry contracts. The statements do not establish a fixed-initial-state device simultaneously performing both tasks on arbitrary unframed input, a physical inverse-edge traversal, or unrestricted state–stack resource optimality. Real-time scalar counters, free-group stack reduction, and generic semigroup realizability are antecedent techniques, not claimed as new.

2 Definitions and the intended model

Let $A \in \mathbb{Z}^{d\times n}$ be an integer counting source on letters $a_1, \dots, a_n$, with each letter costing one. For $b \in \mathbb{Z}^d$ with nonnegative solutions set

$$\ell_A(b) = \min\{\mathbf{1}^{\mathsf{T}} z : z \in \mathbb{N}^n,\ Az = b\}, \quad P_A(b) = \{z \in \mathbb{N}^n : Az = b,\ \mathbf{1}^{\mathsf{T}} z = \ell_A(b)\}.$$

The source is *completion-total* when $Ah = 0$ for some strictly positive integer vector h: any integral solution may then be shifted by a positive multiple of h to become nonnegative. For an optimal fiber P, its active lattice is $L(P) = \langle P - P\rangle_{\mathbb{Z}}$. Choose a vertex $p_0 \in P$ and write $\widehat{P} = P - p_0 \subset L(P)$. Its periodic nerve and quotient are

$$\mathcal{N}(\widehat{P}) = \{F \subset L(P) : F \text{ finite},\ \exists c \in L(P),\ c - F \subset \widehat{P}\}, \qquad Q(P) = |\mathcal{N}(\widehat{P})|/L(P).$$

The quotient is independent of the chosen translation vertex, up to its natural identification. The geometric compression of shortest fibers and the marked difference presentation for $Q(P)$ are established in [2, 3, 4]. In particular, a primitive fiber with m affinely independent vertices has a comparison group F_{m-1}, with displacement onto the rank-$(m-1)$ active lattice given by abelianization; the primitive-simplex specialization is recalled directly in the proof below, while the general marked-quotient framework is used as background. Prior work on semigroup realizations and isolated factorizations supplies related constructions and structural restrictions [5, 6]; it does not by itself assert the exact simultaneous source-and-service contract defined below.

Three types of operational statements must be distinguished. A *whole-language recognizer* answers whether $A\,\mathrm{Parikh}(w) = 0$ for an arbitrary word. A *paired-session comparator* receives fixed-length successive rooted shortest sessions, with externally specified session resets. A *single uninterrupted source trace* has neither those resets nor a separate mode selector. A pushdown realization in one category is not silently promoted to either of the others.

3 An exact facet-and-fiber compiler

Let K have labeled vertices $1, \dots, s$ and facets $F_0, \dots, F_{m-1}$, with $m \geq 1$. Write $v_{ji} = \mathbf{1}_{\{i\in F_j\}}$ and $d_i = \sum_j v_{ji}$. On $(x_i^+, x_i^-)_{i=1}^s, (y_j)_{j=0}^{m-1}, t$, impose

$$x_i^+ - \sum_j v_{ji} y_j + (d_i - 1)t = 0, \tag{1}$$

$$x_i^- - \sum_j (1 - v_{ji}) y_j + (m - d_i - 1)t = 0, \tag{2}$$

$$\sum_j y_j - mt = 1. \tag{3}$$

These are the row equations for a single $(2s+1) \times (2s+m+1)$ integer matrix A_K and one target b.

Theorem 3.1 (Simultaneous source-native realization)**.** *The source A_K is completion-total with the all-ones kernel. Its unit-cost shortest completion length is $s+1$, and the entire optimal Parikh fiber consists of exactly m vectors $p_0, \dots, p_{m-1}$, with*

$$t = 0, \quad y_j = 1,\ y_k = 0\ (k \neq j), \quad x_i^+ = v_{ji}, \quad x_i^- = 1 - v_{ji}.$$

On the designated letters x_i^+, the optimal support complex is precisely K. The fiber is a primitive integral $(m-1)$-simplex in its active lattice. Consequently its native quotient has $\pi_1 Q(P) \cong F_{m-1}$ with natural displacement the abelianization to $\mathbb{Z}^{m-1}$.

Proof. Every row vanishes at the all-ones vector. Adding (1) and (2) gives $x_i^+ + x_i^- = 1 + 2t$, while (3) gives $\sum_j y_j = 1 + mt$. Thus every nonnegative solution has length

$$s(1 + 2t) + (1 + mt) + t = (s + 1) + (2s + m + 1)t.$$

The minimum forces $t = 0$, where the selector y is one-hot, giving exactly the stated p_j. Its positive-letter supports are exactly the facets, and their downward closure is K. The differences $p_j - p_0$ are an integral basis of their span because their selector coordinates $y_1, \ldots, y_{m-1}$ project to the standard basis. For completeness, this primitive-simplex quotient can be seen directly. Translate p_0 to the origin and use the difference basis to identify the fiber with $T = \{0, e_1, \ldots, e_{m-1}\} \subset \mathbb{Z}^{m-1}$. The maximal simplices of its periodic nerve are the translates of $-T$. Two distinct such translates cannot share an edge: two distinct ordered vertex pairs of an affinely independent simplex cannot have the same nonzero difference. Therefore the quotient has one maximal simplex with its m vertices identified and with no further positive-dimensional face identifications. It is homotopy equivalent to a wedge of $m - 1$ circles (a point for $m = 1$). Each free loop lifts with displacement one of the basis differences $p_j - p_0$, so the displacement homomorphism is abelianization. This is precisely the primitive case of the marked-quotient theory cited above. □

4 Whole-language recognition after lossless scalar folding

For a matrix source $A \in \mathbb{Z}^{d \times n}$, target b, and known shortest length L, let

$$M = \max\{1, \max_{j,i} |A_{ji}|, \max_j |b_j|\}, \qquad R > (L + 1)M$$

with integer R. Define scalar letter effects and scalar target

$$\lambda_i = \sum_{j=0}^{d-1} A_{ji} R^j, \qquad \beta = \sum_{j=0}^{d-1} b_j R^j. \tag{4}$$

Theorem 4.1 (Literal preservation of every shortest completion)**.** *For every nonnegative z of length at most L, $\sum_i \lambda_i z_i = \beta$ if and only if $Az = b$. Hence the scalar source has the same shortest length and* exactly the same *optimal Parikh fiber, supports, active lattice, and marked periodic quotient. If $Ah = 0$ with $h > 0$, then also $\lambda h = 0$.*

Proof. Write $e_j = \sum_i A_{ji} z_i - b_j$. Then $|e_j| \le (L + 1)M < R$. The scalar equality is $\sum_j e_j R^j = 0$. Taking residues modulo R forces $e_0 = 0$. Divide by R and repeat; every $e_j = 0$. All original length-L solutions persist and no shorter scalar solutions are introduced. The remaining claims are literal invariance of the fiber and linearity of (4). □

Proposition 4.2 (Real-time one-stack zero-return recognition)**.** *For any integer scalar weights $(\lambda_1, \ldots, \lambda_n)$, the language $\{w : \sum_i \lambda_i \#_{a_i}(w) = 0\}$ has a deterministic real-time one-stack recognizer, using at most one push or pop per input symbol and no empty-input transitions.*

Proof. Set $D = \max(1, \max_i |\lambda_i|)$. Represent the running total uniquely as $Dq + r$ with $0 \le r < D$. Keep r in finite control and q as a signed-unary stack above a bottom marker. On input letter a_i, write $r + \lambda_i = D\delta + r'$ with $0 \le r' < D$. Since $0 \le r < D$ and $|\lambda_i| \le D$, necessarily $\delta \in \{-1, 0, 1\}$. Thus the quotient changes by at most one; make the corresponding push, pop, or no-op and update the remainder to r'. The sign is represented by the stack alphabet $\{\perp, +, -\}$. Only the zero residue and empty signed stack accept. □

Remark 4.3. *For general completion-total vector sources of rational rank at least two, whole-language context-freeness already fails by a standard two-independent-counters reduction, consistent with the group-language literature [7]. The folding theorem changes the matrix and retains only the specified shortest fiber: it does not preserve the entire vector zero-return language. Standard base encoding and one-counter recognition themselves are not claimed as new methods.*

5 A source-faithful protected comparison service

Take the primitive optimal fiber $P = \{p_0, \dots, p_{m-1}\}$ from Theorem 3.1 or its folded scalar source, with minimum length L. Fix the source-native base vertex p_0. One *paired block* consists of two successive genuine shortest source words, each exactly L letters, and an interface reset between them to the same root. Restrict to pairs (p_0, p_j) and (p_j, p_0), $j \geq 1$, assigning symbols X_j and X_j^{-1} respectively. These are comparison labels extracted from source words, not new source input letters. Let $B = \bigcup_{p \in P} \{z \in \mathbb{N}^n : z \leq p\}$ be the finite set of possible shortest-prefix Parikh vectors.

Theorem 5.1 (Protected paired-session comparison)**.** *With this explicitly selected service, there is a deterministic real-time single-stack machine on the* original source alphabet *that checks all session words, updates the freely reduced word in F_{m-1}, and accepts precisely the full paired streams representing the identity. It uses at most $(m+1)|B|+1$ verifier states, stack alphabet of size $1+2(m-1)$, and at most one push or pop on the final letter of a pair, with no empty-input moves. For $m \geq 3$ the service records nonidentity comparisons with zero natural displacement.*

Proof. The first-half verifier stores a prefix Parikh vector $z \in B$ in finite control, rejecting an impossible extension. The completed first half uses $|B|$ states; the second-half states are pairs (i, z) with $i \in \{0, \dots, m-1\}$ and $z \in B$, plus one rejecting sink. Thus the advertised $(m+1)|B|+1$ bound includes both phases and the sink. After L letters, z must equal one unique p_i; remember i and restart the verifier component at zero by the service clock. Check the second half similarly, deriving p_j. If the pair is allowed, read its single $X_k^{\pm 1}$ comparison label at the final source-letter transition. Cancel against the inverse top stack letter or push it; this maintains free reduction inductively. Accept at a completed block boundary if and only if only the bottom marker remains. There is no input alphabet extension and one stack suffices. The chosen group generators correspond to the source's primitive optimal-fiber comparisons. For $m \geq 3$, $[X_1, X_2]$ is nontrivial in the free group but has zero displacement since its exponents sum to zero on every generator. □

Proposition 5.2 (Unbounded comparison requires memory)**.** *For $m \geq 2$, the identity language of complete admissible paired blocks cannot be recognized by a finite-state-only device, although the single-stack construction above recognizes it.*

Proof. Select two fixed genuine paired blocks U, V that produce X_1 and X_1^{-1}. The concatenation $U^a V^b$ represents identity precisely when $a = b$. For distinct a, a', the prefixes U^a and $U^{a'}$ are distinguished by suffix V^a. Thus infinitely many finite-state residuals are needed by the Myhill–Nerode criterion. The proof concerns faithful unbounded pair streams and does not give a minimum number of pushdown states or stack symbols. □

Proposition 5.3 (Sharp semantic restrictions)**.** *Suppose a shortest rooted session has nonzero target displacement b. A concatenation of $k \geq 2$ such sessions has displacement kb, and cannot be one original rooted shortest execution to b. Further, for any completion-total source with $h > 0$, every finite source word is the prefix of some zero-return source word. Thus no*

finite prefix can be reserved exclusively for irreversible comparison-mode routing while still recognizing every zero-return continuation.

Proof. Displacements and lengths add, so the concatenation ends at kb and costs kL, not b in length L. For the prefix claim, if a word has Parikh vector c, choose N with $Nh \geq c$ and append any suffix of Parikh vector $Nh - c$. The completed word has net effect $ANh = 0$. The routing consequence follows immediately. □

Remark 5.4 (Precisely what the construction certifies)**.** *The whole-language counter recognizer and the paired comparison machine are* host-selected alternative services *using one physical stack per run. The theorem does not prove one fixed-start device that concurrently emits both outputs for every unframed raw input, or a physical directed path literally executing formal inverses. Its push and state counts are constructive upper bounds, not general resource minima. These type distinctions are part of the result, not merely implementation commentary.*

6 A seven-letter exact certificate

The folded unit-cost scalar source

$$\lambda = (1, 13, 169, 2041, 2029, 2197, -6450), \qquad \beta = 2211$$

has $\lambda \cdot (1, \ldots, 1) = 0$ and precisely three shortest Parikh vectors of length three:

$$p_0 = (1, 1, 0, 0, 0, 1, 0), \quad p_1 = (1, 0, 1, 1, 0, 0, 0), \quad p_2 = (0, 1, 1, 0, 1, 0).$$

The corresponding source supports are $\{1, 2, 6\}$, $\{1, 3, 4\}$, $\{2, 3, 5\}$; the first three action types induce the boundary of a triangle. The active fiber is a primitive rank-two triangle, and its comparison group is the free group $F(X, Y)$ with the natural rank-two displacement marking. The global zero-return language is recognized by Proposition 4.2 using $D = 6450$ remainder-control states and signed-unary stack. The paired comparison service has $|B| = 19$, at most 77 verifier states and five stack symbols. Four pairs built from the genuine words $a_1a_2a_6$, $a_1a_3a_4$, and $a_2a_3a_5$ yield the reduced word $XYX^{-1}Y^{-1}$, which is nontrivial but has zero displacement. Indeed, the seven-coordinate nonnegative vectors of length at most three number $\sum_{j=0}^{3} \binom{6+j}{j} = 120$. Direct substitution into $\lambda \cdot z = \beta$ gives no solutions of length below three and exactly the three displayed solutions at length three. Each has six distinct orderings; their downsets have union of size 19, and the four allowed ordered types of pairs therefore yield $4 \cdot 6^2 = 144$ six-letter words. The general assertions do not rely on this finite example.

7 Relation to prior results and claims of contribution

The known geometry of completed fibers and their marked quotients is the starting point [1, 2, 3, 4]. Existing semigroup realization and factorization results, as well as free-group word-problem methodology, are substantial neighboring prior art [5, 6, 7]. We claim the explicitly qualified integration of (i) a shared unit-cost source whose visible support complex and shortest fiber are both exact, (ii) literal bounded-length fiber preservation under scalar folding, and (iii) source-letter-only protected paired-session word readout. In particular, our statements are not a classification of unrestricted pushdown implementations or a new theorem on free groups. The hypotheses defining the paired service are essential and are stated in the theorem, rather than left implicit.

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